

SOCAL EDISON SOLAR PANEL INITIATIVE

Background:

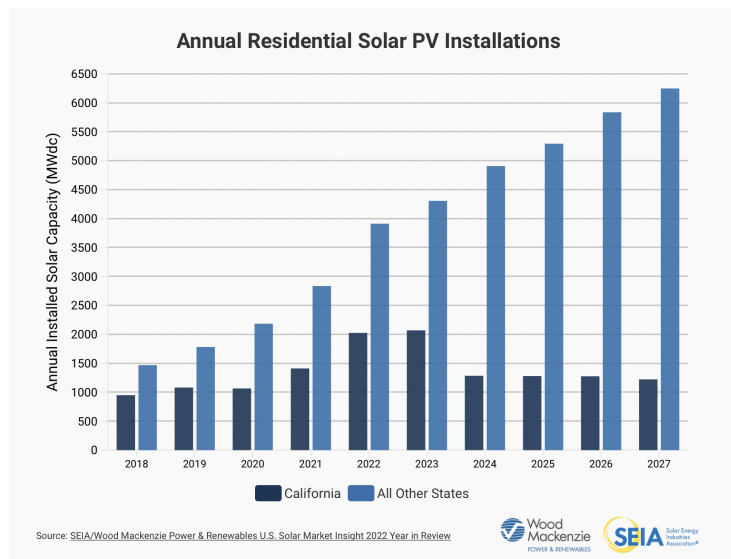
As Southern California Edison continues to improve its grid infrastructure and moves towards a future where 75% of all vehicles in the state will be powered by clean energy sources, the demand for electricity is expected to rise significantly. This presents a critical challenge for SCE to meet this growing demand while achieving its carbon neutrality and clean energy goals. In this context, the deployment of solar panels is a key solution that can provide clean and renewable energy, help reduce the dependence on fossil fuels, and support SCE's efforts to meet its energy demand in a sustainable manner.

With the new Net Billing policy taking place April 14, 2023, new solar customers will no longer be able to take advantage of the previous Net Metering Programs (NEM 1.0, 2.0). The reason for Net Billing is due to efforts of lowering fossil fuels from the energy needed while the sun is down, which encourages solar installation and battery storage. With many of the added costs rising under Net Billing, along with a significantly lower export rate from electricity bill credits, the compensation for extra energy produced reduces by 75%. In turn, the Return on Investment (ROI) from solar jumps from about 6 years all the way to 10 years under the new program.

This big change will make it harder to have new customers invest in solar panels and will decrease the amount of solar panel projects, which will conflict with the growing demand for more clean energy resources and make it harder for customers to deal with the inevitable inflation of electricity costs.

Moreover, companies such as Tesla have had plans to convert all of their EV stations into solar power for the past 6 years, however, they have failed to meet these commitments.

“In 2017, Musk even added that Tesla plans to add solar and batteries to all Supercharger stations and eventually disconnect most of them from the grid. The Supercharger V3 was delayed a few times, but it finally came in 2019. Yet, Tesla didn’t accelerate the deployment of solar and batteries at the charging stations” (Lambert, 2022). During Earth Day, Justin Lange, who oversees Tesla's Superchargers in the mid-Atlantic region, shared intriguing data about the Supercharger network. Among these details, he mentioned that Tesla aimed to make all Supercharger energy 100% renewable by 2021. Although some Supercharger stations currently use solar power, the majority



still rely on the grid and receive electricity from local utility companies, which means that previous commitments have not been fulfilled (Lambert, 2021).

As access to solar panels becomes more challenging and electricity demand continues to rise, what can be done to effectively increase the availability of renewable energy resources and support SCE's goal for a more sustainable future?

Problem Statement:

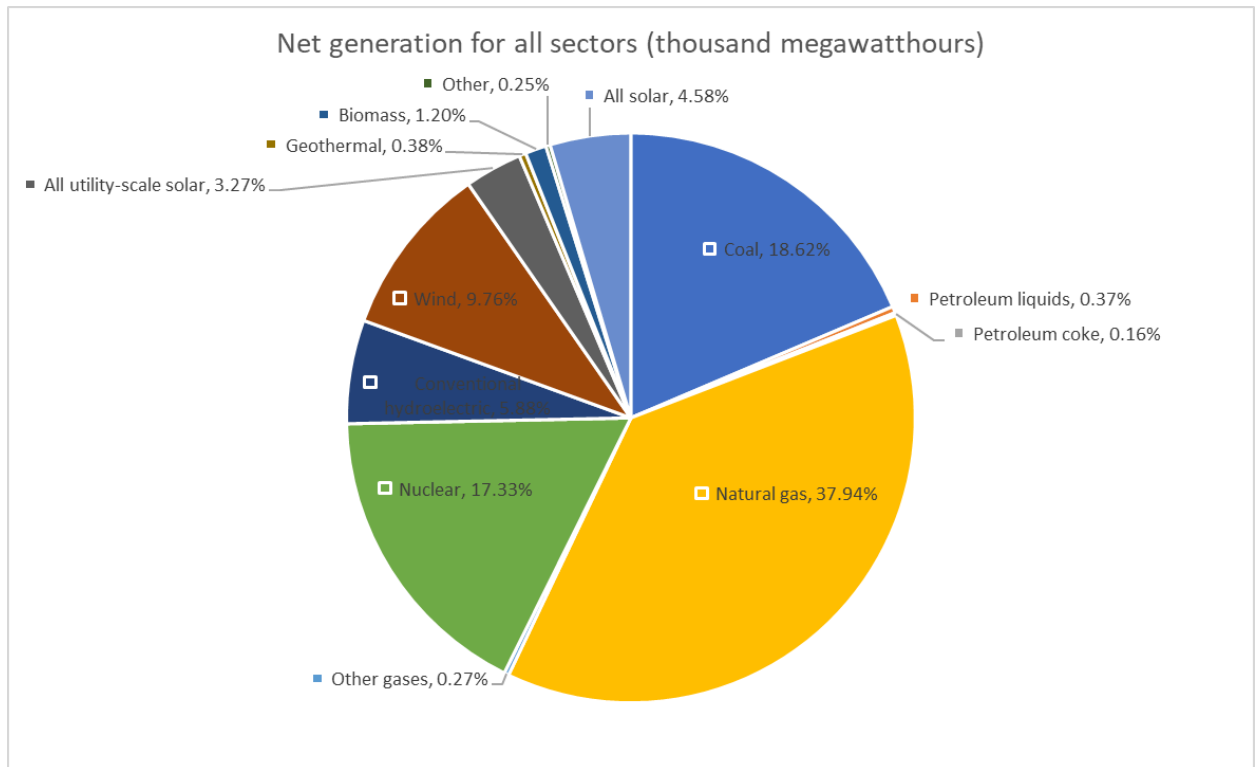
As Southern California Edison (SCE) works to meet the growing demand for electricity in a sustainable manner and achieve its clean energy goals, it faces the challenge of adapting to the new Net Billing policy, which may discourage new customers from investing in solar panels due to increased costs and longer Return of Investment (ROI). This could lead to fewer solar projects, making it essential for SCE to redirect solar efforts toward electric vehicle (EV) infrastructure. Many companies, like Tesla, have not met their commitments to convert all their EV stations into solar power, thus presenting an opportunity for SCE to support the development of solar-powered EV infrastructure to ensure grid reliability, affordability, and the achievement of clean energy targets. Furthermore, addressing the problem of effectively integrating solar power generated from electric vehicle charging stations into our overall energy system is crucial for optimizing renewable energy use and promoting sustainability.

Objective:

The objective is for SCE to develop solar-powered EV infrastructure to ensure reliable and affordable clean energy, taking advantage of the opportunity presented by companies like Tesla not meeting their commitments, starting with LA County. This will be done by integrating solar power from EV charging stations effectively into the overall energy system to promote sustainability.

I The Crucial Role of Solar Panels in Sustainable Energy

Today's modern world makes use of an abundant amount of electricity. Through the years 1990 - 2021, the county of Los Angeles alone was reported to have a total usage of 2156592.470931 kWh (GWh) of electricity which is 25.55% of all electricity consumed throughout California within the years 1990 - 2021. To keep up with the demand for high electricity consumption rate, the United States generates electricity using various resources. The following diagram depicts the variety of resources used to produce electricity:



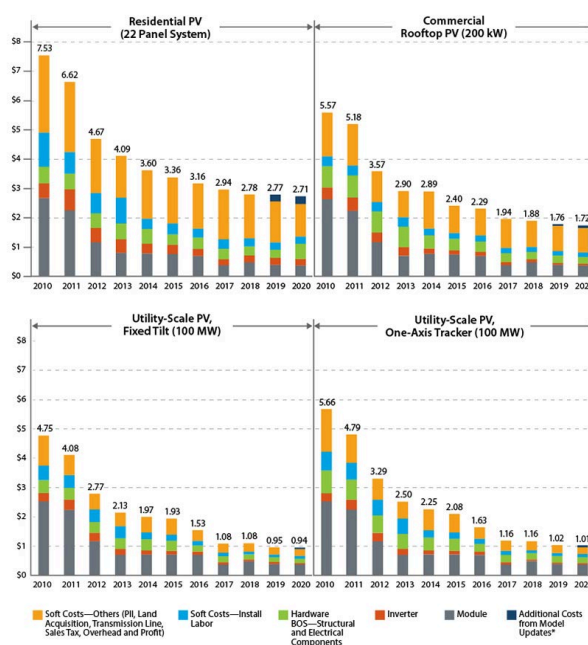
The three most commonly used resources reported are natural gas, coal, and nuclear power. Although the three are reliable sources of acquiring energy, they have significant drawbacks that require careful consideration when making energy decisions.

The energy sources are non-renewable in that “their supplies are limited to the amounts that we can mine or extract from the earth. Coal, natural gas, and petroleum formed over thousands of years from the buried remains of ancient sea plants and animals that lived millions of years ago.” (3). Since these resources are finite, their availability will eventually decline and their prices will increase as demand continues to grow, making them limited energy sources in the long run. William Walsh, an executive at Southern California Edison discussed the increased energy sources prices with LA Times and reported how “his company had filed for a June 1 rate increase, totaling \$595.6 million. The increase, which would be 4.4% for the average customer, is still pending approval by the CPUC.” (5). As discussed as availability decreases, the prices increases which in return also increases the monthly electricity bill for customers.

Additionally the energy sources also have significant impact on the environment as the process to derive them from the Earth contributes to air pollution, water pollution, climate change, and other harms to the planet. The Union of Concerned Scientists has drawn attention to the negative effects that non-renewable energy sources, particularly natural gas, have on the environment. They found that the extraction and production of natural gas cause contamination of the air, water, and the release of methane, a strong greenhouse gas. Natural gas consumption, which contributes to climate change and other environmental problems, exacerbates these challenges.

To keep up with the growing demand for electricity, it's becoming increasingly important to find renewable and sustainable energy sources that are also environmentally friendly. Solar energy is a type of energy source that fits these criteria, as it converts sunlight into electricity that can be used in both residential and non-residential sectors. Solar energy is generated using photovoltaic (PV) cells placed in panels which absorb energy from the Sun and then convert it into electricity using semiconducting materials. This process can then be repeated everyday as the Sun has limitless energy for another 5 billion years. Additionally, unlike the most commonly used resources to generate electricity the use of solar panels is environment friendly as it doesn't produce any greenhouse gases in the process.

While solar energy is a promising renewable and sustainable energy source, there are some drawbacks to consider. The installation of solar panels can be expensive for many homeowners or businesses to be able to afford. The following diagram below depicts the total costs of installing solar panels for residential rooftop, commercial rooftop, and utility-scale ground-mount systems:



According to the National Renewable Energy Laboratory, the installed cost of residential solar photovoltaic systems in the United States was \$7.53 per watt in 2010. This would mean that a 5-kilowatt solar panel system, which is the average size for a residential installation, could cost around \$37,650. In 2020 however, the price in 2020 was decreased to \$2.71 per watt, which would mean that the average 5-kilowatt solar panel residential installation could cost around \$13,550. Although the cost of installation has decreased significantly in recent years, it can still be a significant barrier to adoption for many homeowners. However, it's important to note that the long-term cost savings from reduced energy bills and potential earnings from selling excess energy back to the grid can make solar energy a financially beneficial investment over time.

II Regulations related to Solar

The following are relevant regulations in California related to Solar:

- California Solar Initiative (CSI) - Provides financial incentives based on the size of the system and other factors to customers who install solar panels on their homes or businesses
- Net Energy Metering (NEM) - Allows residential and commercial customers with solar or other renewable energy systems to receive a credit on their utility bill for any excess electricity they generate and feed back into the grid
- The California Building Code - Builders must install electrical wiring and other infrastructure that will allow for the installation of solar panels in the future
- Green Building Standards Code (CALGreen) - Requires new residential and commercial buildings to meet provisions for solar energy systems
- Renewable Portfolio Standard (RPS) - Requires utilities to increase their use of renewable energy sources
- 2035 electric car mandate - All new passenger cars and trucks sold in California are required to be zero-emissions vehicles (ZEVs) by the year 2035

III Research Findings

An experiment was done to test out a potential implementation plan for solar electric vehicle stations (Affonso and Kezunovic, 2019). The idea was to put solar panels on the roof of the building or make a special parking spot with solar panels called a "solar carport" that would give energy to the charging station and building. However, the electric grid will still be connected to both the building and the charging station. If the solar panels run out of energy, then the electric grid will provide both structures energy.



Figure 1. System under study.

The figure above shows the building and charging station being connected to each other. The solar panels (on the left) at the station will power the charging station and the building. If there is not enough energy from the solar panels, the electric grid will provide energy to both structures.

The authors used a special computer program and advanced statistics to help them estimate how much solar power the solar panels could make, how much power the electric cars would need, and how much electricity the building would use. Here are some conclusions made from this study:

1. When the charging time of electric vehicles aligns with the solar energy production period, solar power can fulfill approximately 65.47% of the charging needs on average.
2. Although electric vehicles may not always be charged precisely when solar panels generate the highest amount of electricity, the solar charging station can still contribute to a 6.5% reduction in the total monthly electricity cost.
3. In the winter months, the monthly electricity expense can be lowered by about 5.5%, a satisfactory outcome.

All in all, they found that using solar power for electric car charging stations would be feasible and positive in the long run. These stations can make enough power for the cars and the building, and it can even save money in the long run. By integrating the solar panels with the building and the charging station, a more sustainable and efficient energy system will be created. Additionally, concerns about insufficient energy or sunlight would be alleviated, as the electric grid serves as a backup to ensure a reliable power supply.

IV Implementation

To set up a solar energy system that powers both a building and an electric vehicle (EV) charging station, a series of crucial steps and components are involved. The process starts with a site assessment to evaluate the location for solar panel installation, taking into account space availability, shading, and optimal panel orientation (Wolf, n.d.). Based on the assessment, solar panels can be installed on rooftops, canopies, or ground-mounted systems, depending on the specific situation.

Solar panels generate direct current (DC) electricity, which must be converted to alternating current (AC) for use in the building and EV chargers. In DC, electricity is maintained at constant voltage in one direction. In AC, electricity flows in both directions in the circuit as the voltage changes from positive to negative (“Solar”, n.d.). When electricity flows from both directions, it can flow through the EV charging station and building, thus generating power for both. An inverter accomplishes this conversion and connects to both the main electrical panel and the EV charging station, enabling solar-generated electricity to power both. To guarantee a consistent power supply, the system connects to the electric grid.

Optionally, a monitoring system could be implemented to track solar energy generation, energy consumption, and charging station usage to optimize the system, identify potential issues, and facilitate regular maintenance scheduling.

By following this approach, a solar-powered system that powers both a building and an EV charging station can be successfully established, leading to a more environmentally friendly and efficient energy solution that reduces emissions and fosters the shift to cleaner transportation options.

V Estimated Cost:

According to the U.S. Department of Transportation, for every 36.92 miles that is driven, a home EV charger would use around 11.81 kWh per day to charge the car to replenish the range driven (“How”,

n.d.). Additionally, the daily driving distance for most drivers is around 29.5 miles (U.S. Department of Transportation, Federal Highway Administration, 2020). A 6 kW solar carport can generate 9,000 kWh per year, or 24.66 kWh per day (David, 2023). According to EnergySage, a solar carport has an average cost of \$3.31 per watt. Assuming that 10 electric vehicles get charged at a private EV station (if more vehicles charge, then the electric grid can provide energy), the costs involved can be broken down as follows:

Demand of Energy:

- Since charging 36.92 miles uses 11.81 kWh, then 29.5 miles uses 9.42 kWh.

$$(29.5/36.92)*11.81 = \mathbf{9.43 \text{ kWh}}$$

Power Generated per 1 kW solar carport:

- Amount of power generated for 1 kW solar carport:

$$24.66 \text{ kWh per day} / 6 \text{ kW solar carport} = \mathbf{4.11 \text{ kWh}}$$

Legend

- x = amt of cars
- y = Solar Carport size (kW) needed to meet demand
- $9.43x$ = kWh demanded per day
- 1 kW solar carport = 4.11 kWh
- 1 kW = \$3,310

Solar Carport Cost Formula:

- $9.43x/4.11 = y$
- $3310y$ = Cost of installation

Solar Carport Size	Typical Project Cost	30% Federal Tax Credit	Net Project Cost
5 kW	\$16,550	\$4,965	\$11,585
6 kW	\$19,860	\$5,958	\$13,902
7 kW	\$23,170	\$6,951	\$16,219
8 kW	\$26,840	\$7,944	\$18,536
9 kW	\$29,790	\$8,937	\$20,853
10 kW	\$33,100	\$9,930	\$23,170

The chart above shows the cost of a solar carport of different sizes, including the 30% tax credit (David, 2023)

A solar inverter, which converts DC electricity from solar panels to AC electricity for use in buildings and charging stations, will cost between 9% to 16% based on the chart below (for SMA and ABB), as well as our estimated cost for one solar carport. When it comes to selecting the most suitable technology for solar EV stations and buildings, it depends on various factors, such as the size of the installation, budget, and energy goals. However, for larger-scale projects, such as commercial buildings, microinverters or power optimizers may be the better choice, as they offer greater flexibility and better energy production.

With micro inverters, each solar panel has its own inverter, which provides greater system flexibility and better energy production, especially if there are shading issues on the roof. Power optimizers, on the other hand, optimize energy production for each panel, allowing for greater efficiency and longer panel lifespan. Overall, the best technology will depend on the specific needs of the commercial building and its EV charging stations.

Manufacturer	Type	Model	Cost	% Total Cost of Average Installation (5.6kW, \$16,408 total)
SMA	String	SunnyBoy 5.0-US	\$1500	9%
ABB	String	PVI-5000	\$1580	9%
Solar Edge	Power Optimizer	Inverter/Power Optimizer combo (17 total)	\$2,323	14%
Enphase Energy	MicroInverter	M250 microinverters (20 total)	\$2,680	16%

The chart above shows different solar inverter models and their costs (Austin, 2017).

Inverter Formula

$$\rightarrow 0.15 \times \text{Cost of installation} = \text{Cost of Inverter}$$

The cost of the (already established) electric grid will be based on the amount of electricity used. In this case, we estimate a very small percentage of electricity from the grid to be used.

Overall, the estimated cost of implementing a solar carport that will likely run a commercial building and the EV station on solar energy will be the sum of the inverter and the size of the solar carport installed.

VI Why would private companies invest in this new system?

Companies have struggled to achieve their sustainability goals, particularly when it comes to transitioning EV stations to solar power. The challenges arise due to inconsistent solar generation, and the high upfront costs associated with installation. Companies may be hesitant to invest substantial amounts of money into a system that has a low energy output during periods of low solar generation. However, an integrated solar EV station plan that connects the solar panels to building infrastructure with backup from the electric grid addresses these problems and proves to be a more promising and efficient solution compared to the previous standalone solar panels.

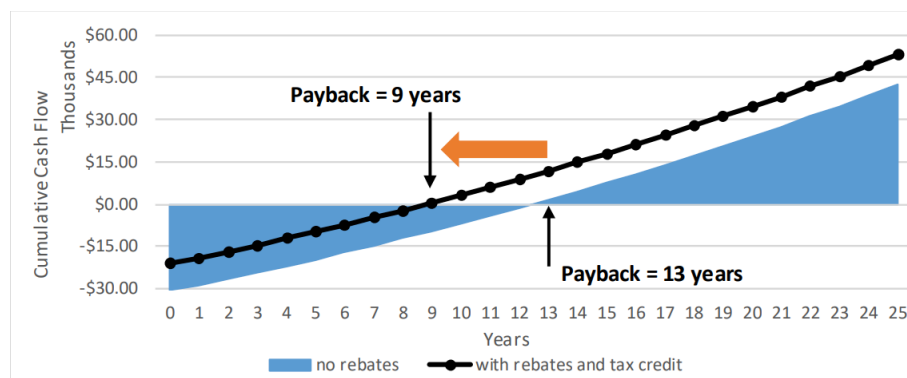
Integrating solar EV charging stations with building infrastructure and utilizing the electric grid as a backup supply offers several benefits over merely using standalone solar panels. Connecting solar EV stations to buildings, it maximizes the use of space and resources, as solar-generated energy can power both the building and charge electric vehicles. This interconnected system results in more effective energy management.

Additionally, the electric grid's backup ensures continuous and dependable power availability. Solar panels' energy output depends on weather conditions and the time of day, which can be inconsistent. By linking solar EV stations to the grid, companies can provide uninterrupted charging services for electric vehicles, even during periods of limited solar generation or increased energy consumption. This dependability is vital for encouraging the widespread adoption of electric vehicles and maintaining customer satisfaction.

Moreover, the elimination of the Net Metering Program has caused companies to perceive the excess energy generated by their solar panels as a burden rather than an opportunity. In the absence of compensation for the excess energy, it is easy to overlook the potential benefits of this surplus power. However, companies could capitalize on this unused energy by utilizing it to power their buildings. By implementing energy storage systems or upgrading their electrical infrastructure, businesses can store and redistribute the excess energy generated by their solar panels. This allows them to reduce their reliance on traditional energy sources and increase their energy independence while simultaneously reducing their carbon footprint.

Some may contend that combining solar EV charging stations with building infrastructure and using the electric grid as a backup is more complicated and expensive than merely employing standalone solar panels. The installation process might demand extra planning, engineering, and collaboration with local utilities, possibly raising initial costs and extending the time needed for implementation. Though the integration of solar EV charging stations with buildings and the electric grid may indeed involve more initial complexities and costs, the enduring advantages surpass these challenges. The effective utilization of space and resources, along with the dependability provided by grid connectivity, delivers a more sustainable and reliable energy solution. In the long run, the interconnected system can lead to decreased energy expenses and increased customer satisfaction, justifying the initial investment.

Moreover, companies investing in solar-powered electric vehicle (EV) charging stations can reap substantial long-term financial benefits, with government incentives and subsidies speeding up the process. One notable example is the Solar Investment Tax Credit (ITC), which offers homeowners and commercial business owners a 30% tax credit (as of 2023) on the costs associated with installing solar panel systems ("Federal").



The illustration above shows that the break-even point is accelerated by 4 years with the aid of government rebates and tax credit (Affonso and Kezunovic, 2019).

Additionally, adopting a comprehensive solar EV charging station plan can strengthen a company's brand image and reputation by emphasizing its commitment to sustainability, thereby resonating with eco-conscious consumers and stakeholders. Furthermore, companies that lead the way in merging solar systems with building infrastructure and implementing electric grids as backup plans can showcase their innovative spirit and dedication to sustainable energy solutions. This pioneering approach not only enhances their reputation but also inspires other companies to join the movement, ultimately fostering a more significant positive impact on the environment and overall energy sustainability. Moreover, this plan provides companies with a sense of security, reliability, and less risk of failure with the electric grid as a backup.



Sources: Bain Sustainability and Change survey (n=301); Bain risk history survey (n=318)

The figure above shows that the majority of companies have failed to achieve their sustainability goals. (“98%).”

Moreover, based on the previous study analyzed in section II of this paper, there will about 65.47% of this system will use solar energy. Using this information, we can calculate the decrease in overall carbon emissions released into the air:

Carbon emissions (metric tons of CO₂e) = (Electricity consumption in kWh) x (Carbon intensity of electricity in kg CO₂e/kWh) / 1,000

According to the California Energy Commission, the carbon intensity of electricity in Southern California was 0.314 kg CO₂e per kWh in 2020.

Based on this information, the formula for carbon emissions in Southern California will be:

Carbon emissions (metric tons of CO₂e) = (Electricity consumption in kWh) x (0.314) / 1,000

Based on this formula, for every 1 kWh, 0.00318 metric tons of CO₂e is emitted into the air when using the electric grid. When using this new solar system, there will be 0.0011 metric tons of CO₂ emitted into the air for every kWh.

VII What Would Make Solar EV Stations Work the Best?

The optimal conditions for a solar EV station to maximize its energy production and efficiency depend on several factors. First, the station's solar panels should be oriented to face the sun as directly as possible, ideally at an angle that maximizes solar radiation during peak sunlight hours. The optimal temperature for a solar panel is 59 - 95 degrees Fahrenheit. This is why we will focus on areas around this range of temperature

The location of the station is also crucial, as it should be situated in an area with high levels of sunlight and minimal shading from surrounding trees, buildings, or other structures. This is why we will target areas that have the least amount of obstructions from receiving sunlight.

In addition, the type and quality of the solar panels used in the station can impact its energy production. High-efficiency panels that can capture more sunlight and convert it into usable energy are preferable, as they can generate more electricity with a smaller surface area. Because we are working with EV stations in busy areas, it is important to have the highest quality solar panels.

Finally, the station's energy storage and management systems can play a crucial role in optimizing its energy production. By using high-capacity batteries and sophisticated energy management software, a solar EV station can store excess energy during periods of high sunlight and distribute it during periods of low sunlight, allowing it to operate more efficiently and effectively. Overall, the best conditions for a solar EV station to maximize its energy production and efficiency involve a combination of optimal panel orientation and placement, high-quality panels, and sophisticated energy storage and management systems.

VIII How the Previous Study Can Be Used in Southern California

Because the study that was examined was conducted in Austin, Texas, it is important to consider whether the benefits of installing solar panels over the electric vehicle (EV) charging stations that are connected to buildings could apply to other regions with high adoption rates of EVs, such as Southern California, where Southern California Edison is the primary energy provider. The study suggests that this approach not only generates clean energy but also reduces strain on the power grid during peak hours. In the following text, we will delve into the details of the Austin study and examine the potential, logistics, and challenges of applying this model in Southern California. We will be starting off by using Los Angeles County first since it is one of the most energy-consuming areas in not only California but the United States.

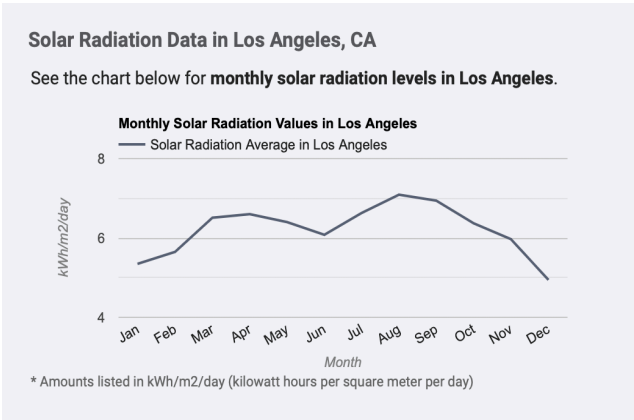
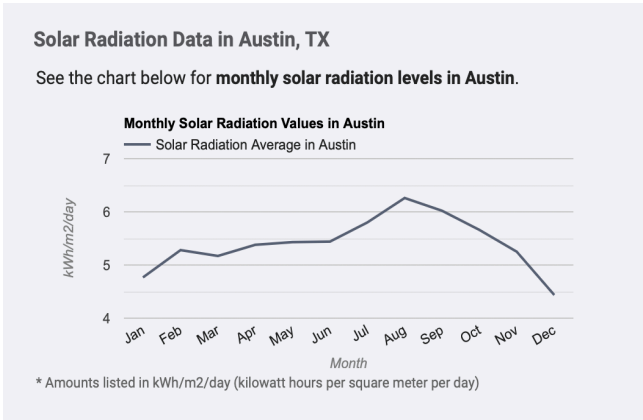
IX Why Los Angeles County?

Los Angeles is one of the most energy-consuming places in the United States, which makes it even more important to explore sustainable energy solutions like solar EV stations. It is also an ideal

location for solar EV stations due to its abundant sunlight, optimal conditions for solar power, and high solar capacity. With an average of 284 sunny days per year, there is ample opportunity to generate solar power, which could be used to charge electric vehicles. The warm climate with low humidity in Los Angeles is also ideal for solar panel efficiency. Solar panels perform better in cooler temperatures, and the lower humidity levels help reduce the buildup of dust and dirt on the panels, which can decrease their effectiveness over time.

Furthermore, California leads the U.S. in solar energy production, and as of currently, Los Angeles has a solar capacity of 1,096 MW, making it the third highest county in the state. Installing solar EV stations in Los Angeles could help reduce greenhouse gas emissions and promote the adoption of electric vehicles, which is particularly important for a city that has struggled with air pollution for decades. Additionally, the city has set ambitious renewable energy goals, including a target of 100% clean energy by 2035, and solar EV stations could help Los Angeles achieve these goals.

Austin, Texas vs. Los Angeles County

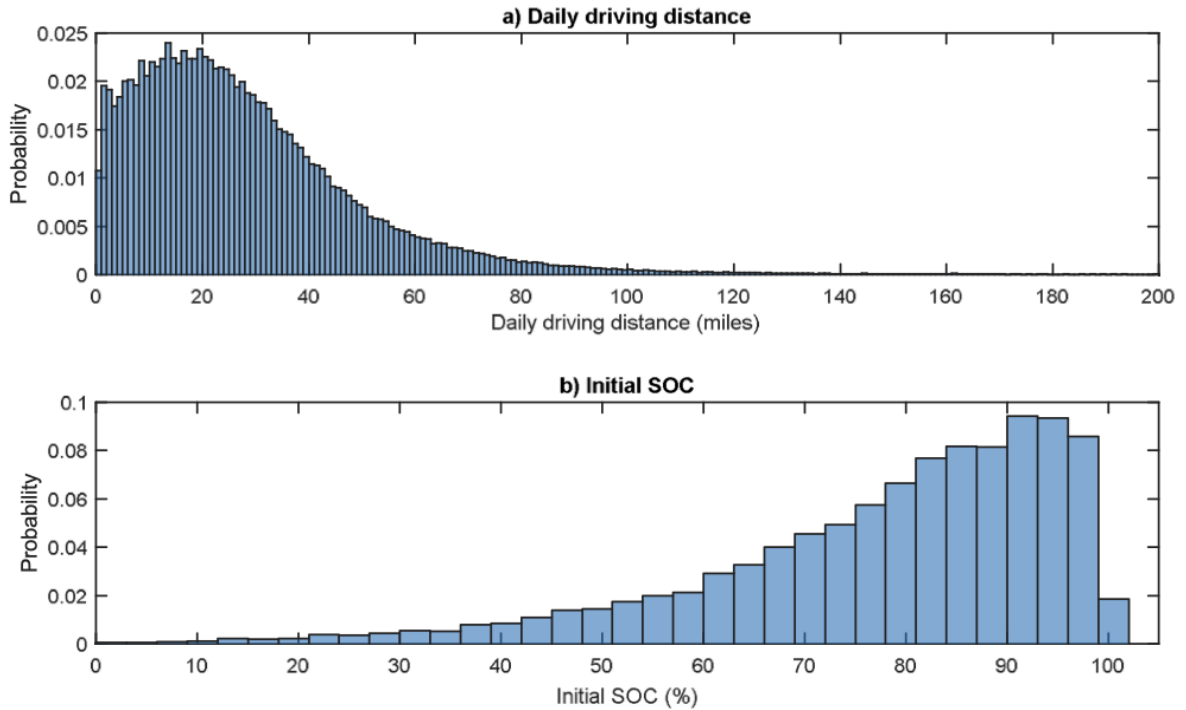


If Los Angeles has higher maximum radiation than Austin, it suggests that there is more potential for solar power in Los Angeles. This means that solar panels installed in Los Angeles could potentially generate more energy than those installed in Austin, all other factors being equal. The other factors that will be discussed are panel efficiency, orientation, and shading, as well as policies, incentives, and regulations.

X Private or Public EV Stations?

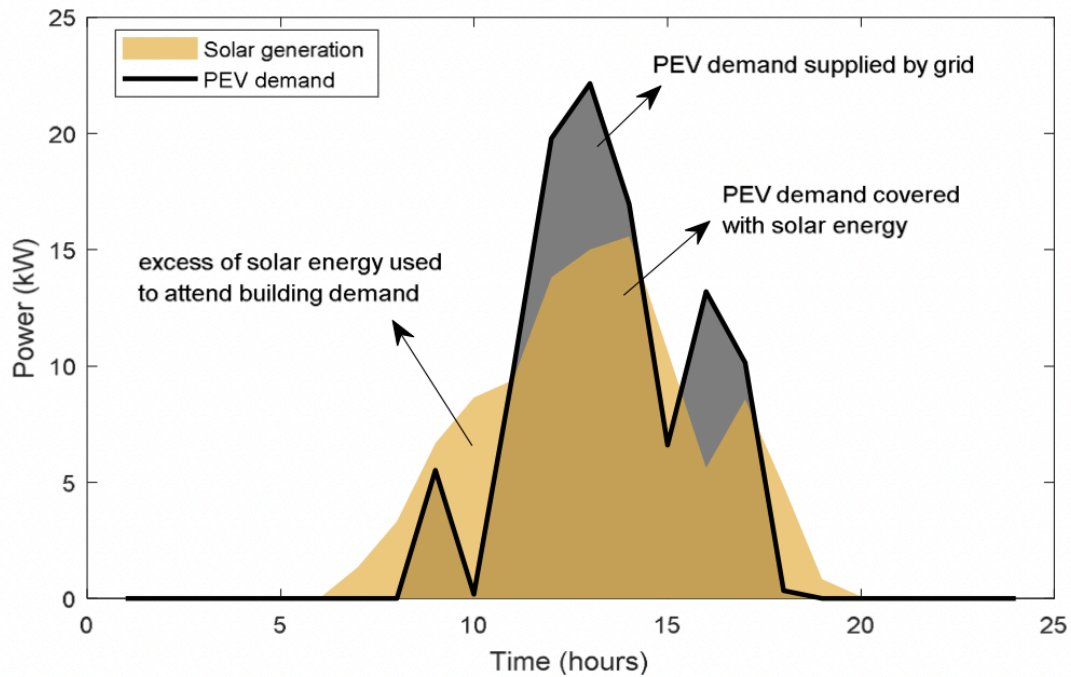
To start off, private buildings will be targeted first to install solar EV stations. Private buildings are often located in areas with high population density, which makes them ideal locations for EV charging stations. Installing solar-powered EV charging stations in these areas can help promote the adoption of electric vehicles, reduce greenhouse gas emissions, and support the growth of clean energy infrastructure. By targeting private EV charging stations near private buildings, we can leverage the potential for increased adoption of clean energy solutions in densely populated areas,

and expand to other locations in the future. This will help see if this would work if placed on a larger scale, which would be a part of our ten-year plan.



The SOC data and daily driving distance can be used to estimate the demand for EV charging during the day. Based on the SOC data presented in the study, we know that most EVs have an initial SOC between 70-90%, indicating that they do not need a full charge. Additionally, the daily driving distance for most drivers is around 29.5 miles (U.S. Department of Transportation, Federal Highway Administration, 2020), which means that they do not require a full charge to complete their daily driving needs.

Considering this information, it becomes clear that EV charging during the day is more efficient and convenient for most drivers. By charging during the day, EVs can be charged to a sufficient level to meet drivers' daily needs without requiring a full charge. This reduces the load on the grid and ensures that excess energy produced by solar EV stations can be used to meet other building demands nearby. Therefore, incorporating SOC and daily driving distance into the equation can help to determine the most efficient and effective way to deploy solar EV stations in a given area.



The study evaluated the feasibility of using a PV system to supply the electricity demand of a fleet of PEVs in a building. The study used aggregated data for the PEV demand and solar energy generation to evaluate the potential of this approach. Aggregated data means that the study analyzed the combined energy demand of the entire fleet of PEVs, and the combined energy production from the PV system.

This approach can be applied to different locations in Los Angeles County with varying numbers of units per EV station. The feasibility of using a PV system to supply the electricity demand of a fleet of PEVs in a building will depend on factors such as the size of the fleet, the number of units per EV station, and the availability of solar energy.

To use this approach, data for the electricity demand of the fleet of PEVs in a building and the solar energy generation potential for the location can be collected. The data can then be used to simulate the energy demand and generation using the equation mentioned in the question:

$DC = 100 \times \% \sum [Pt_{PEV, PV} / P_{PEV}] / N$, where $Pt_{PEV, PV}$ is the aggregated tPEV demand at time step t supplied by solar generation, P_{PEV} is the aggregated PEV demand at time t, and N is the number of intervals in simulation.

The resulting percentage represents the portion of the aggregated PEV demand that can be covered by solar energy generated from the PV system. This information can be used to evaluate the feasibility of using a PV system to supply the electricity demand of the fleet of PEVs in a building. If the energy demand of the fleet of PEVs is high, a larger PV system may be needed to generate enough solar energy to meet the demand. If the number of units per EV station is higher, the energy

demand per station will also be higher, and a larger PV system may be needed. On the other hand, if the solar energy potential for the location is high, a smaller PV system may be sufficient to meet the energy demand of the fleet of PEVs.

The approach used in this study can be applied to different locations in Los Angeles County with varying numbers of units per EV station. By collecting data on the electricity demand of the fleet of PEVs and the solar energy potential for the location, the feasibility of using a PV system to supply the electricity demand can be evaluated using the equation mentioned in the question. This approach can help to reduce the carbon footprint of the transportation sector and promote the use of renewable energy sources.

XI Why This is Better Than Using Solar Batteries

Electric vehicles (EVs) are growing in popularity as the world moves toward more sustainable energy sources. However, it is nevertheless challenging to charge these vehicles on the go. Although battery technology has advanced significantly, its capacity is still constrained, and charging times are still lengthy. On the other hand, installing solar EV charging stations next to structures has a number of benefits.

Using solar panels to power electric vehicle (EV) charging stations and connecting them to buildings is more efficient than using standalone solar batteries to power EV stations. The reason for this is that solar panels generate energy during the day when the sun is shining and the demand for electricity is high, especially in commercial areas where EV charging stations are located. By connecting the solar panels to the building's electrical grid, excess energy generated by the solar panels during the day can be used to power the building's other electrical needs, reducing reliance on the electrical grid and saving money.

CNBC's article "Solar batteries—are they really green?" discusses the usage of solar battery and describes how "Some businesses are being sold battery systems to store energy, but not necessarily from solar or other renewable resources. In many cases, such installations are designed purely to profit from storing grid energy at night, then selling power back to the grid during the day when electricity prices are higher." (Tamminen 3). The use of battery storage systems to store extra solar energy, notably in residential and commercial buildings, is covered in the article. Battery storage may be a smart concept for residential buildings with adequate room to produce excess solar energy, but commercial buildings often use more energy than they can produce, making it impossible for them to use excess solar energy to charge their batteries. Instead, they would be forced to rely on the grid to charge their batteries, which could result in more damaging emissions and a greater reliance on fossil fuels. Likewise, batteries also lose 10% of their power during the charging and discharging process, and battery installations have been shown to increase greenhouse gas emissions.

In contrast, using standalone solar batteries to power EV stations means that the solar panels generate energy during the day and store it in batteries, which are then used to power EV stations at night or during periods of low sunlight. However, this approach is less efficient because the batteries have limited capacity and may not be able to store enough energy to power EV stations for an

extended period of time, particularly during peak demand periods. Additionally, standalone solar batteries are more expensive and require maintenance, which can add to the overall cost of the system.

Additionally, the usage of batteries for EV stations isn't much cost-effective and reliant compared to the grid system. Forbes mentions that: "if batteries in the charging station are nearly full, any solar power produced at that time will simply be thrown away. That means it won't offset the use of fossil power on the grid, and it's also an economic loss." (Templeton 3). In the article, it is discussed that off-grid EV charging stations with batteries may occasionally be more cost-effective, although this strategy has a number of disadvantages. The price of the batteries themselves, which can be fairly expensive, is one of the main problems. Additionally, improper battery sizing could prevent them from meeting the peak EV charging requirements or cause them to accumulate extra solar power that cannot be used when it is needed. This might lead to energy waste and a bigger carbon footprint. Despite the fact that batteries have their uses, grid-tied EV charging stations are still the most effective and environmentally beneficial alternative in the majority of areas.

XII EV Level 2 EVSE Locations

Placing solar carports for electric vehicle (EV) stations is an innovative way to promote sustainable energy and transportation. However, it can be expensive to install these stations across a wide area. To address this issue, it makes sense to target areas with the lowest number of Level 2 EV charging units and the lowest amount of EV sales. By targeting these areas, we can experiment with solar carports and gauge their effectiveness without incurring a significant expense.

This targeted approach is also a safer way to test the waters before investing more resources into widespread implementation. By focusing on areas where EV usage is low, we can ensure that the demand for solar carports is manageable, and we can assess the feasibility of the project without committing to a large-scale deployment. This approach also enables us to reach areas that are currently underserved in terms of EV charging infrastructure, which can help to promote greater adoption of electric vehicles in those regions.

Furthermore, by utilizing solar carports for EV stations, we can help to reduce greenhouse gas emissions and promote clean energy. Solar carports are designed to capture energy from the sun and convert it into electricity, which can be used to power EV charging stations. This energy source is renewable and emissions-free, making it an ideal way to power electric vehicles.

Targeting areas with low EV sales could be a reasonable approach for testing the effectiveness of solar-powered EV stations. The idea is to start with a small-scale project that can be easily managed and monitored. By selecting a low-risk area with low EV sales, the impact of the project on the local power grid and the community can be carefully evaluated. This can help determine whether the solar-powered EV stations are feasible and cost-effective in the long run, as well as identify potential issues that need to be addressed before scaling up the project.

In addition, targeting areas with low EV sales can also help promote EV adoption by providing convenient and affordable charging options. This can help increase demand for EVs and reduce the reliance on fossil fuels, which is important for achieving sustainability goals.

Evidently, targeting areas with the lowest number of Level 2 EV charging units and the lowest amount of EV sales is a practical and effective approach to implementing solar carports for EV stations. By doing so, we can experiment with this technology in a controlled manner, promote sustainable energy, and encourage greater adoption of electric vehicles in areas that are currently underserved.

XIII Overall Goal for LA County

As we continue to experiment with solar carports for EV stations in areas with low EV usage and limited charging infrastructure, our ultimate goal should be to expand to locations that consume the highest energy. In particular, densely populated areas and urban centers with high traffic volumes are the most significant sources of greenhouse gas emissions and air pollution. By targeting these locations, we can make a significant impact on the environment while also reducing costs for EV owners and operators.

The environmental benefits of transitioning to solar-powered EV charging stations are numerous. Not only does this technology generate clean energy, but it also reduces our reliance on fossil fuels and decreases our carbon footprint. Furthermore, it helps to improve air quality in areas that are prone to high levels of pollution, which can have a significant impact on public health. By installing solar carports in these areas, we can promote the use of electric vehicles, reduce emissions, and create a cleaner, healthier environment for everyone.

In addition to the environmental benefits, transitioning to solar-powered EV charging stations can also lead to significant cost savings. EV owners can enjoy lower fuel costs compared to traditional gasoline-powered vehicles, while operators of charging stations can save on energy expenses by harnessing solar power. Furthermore, by strategically placing solar carports in areas with high traffic volumes, we can reduce the need for costly grid upgrades, which can help to keep energy costs low and increase access to charging infrastructure.

Expanding the use of solar carports for EV stations in high-energy areas is a critical step towards achieving a more sustainable, cost-effective transportation system. By leveraging solar power and promoting the adoption of electric vehicles, we can reduce our reliance on fossil fuels, improve air quality, and create a cleaner, healthier environment for everyone. Furthermore, this transition can lead to significant cost savings for both EV owners and operators of charging stations, making sustainable transportation more accessible and affordable for all.

IV Greenhouse Gas Emissions Results

According to the California Energy Commission, the carbon intensity of electricity in Southern California was 0.314 kg CO₂e per kWh in 2020.

Carbon emissions (metric tons of CO₂e) = (Electricity consumption in kWh) x (Carbon intensity of electricity in kg CO₂e/kWh) / 1,000

Electric Grid:

$$1 \text{ kWh} = 1 * 0.314 / 1000 = \mathbf{0.000314 \text{ CO}_2/\text{kWh}}$$

New Solar System:

$$0.000314 * 0.65 = 0.0002041$$

$$0.000314 - 0.0002041 = \mathbf{0.0001099 \text{ CO}_2/\text{kWh}}$$

VI Proposal of Using European Commission's Joint Research Center Photovoltaic Geographical Information System

The JRC Photovoltaic Geographical Information System tool was developed to estimate the performance of solar systems using worldwide solar data.

To get started, make sure the “Grid Connected” tab is chosen, as our panels will be connected to the grid. Then, choose the location of the desired solar panel on the map or input the coordinates in the box. From there, select PVGIS-NSRDB as the solar radiation database. This database, which stands for National Solar Radiation Database, consists of satellite-derived data centered in the United States, so it is the better option here. The PV technology here should be set to ‘Crystalline’. The other two options, CIS and CdTe, are thin-film panels. Because crystalline panels are more durable and have roughly double the energy efficiency of thin-film panels (Clancy, 2022), crystalline will be our panel of choice. We will put 300 kWp for the installed peak power, as it is the number we established earlier on how much a commercial building uses. The system loss percent refers to energy loss during a panel’s operation due to various factors such as shading, maintenance issues, dirt, etc. A good system loss percentage is usually between 10-15%, so we will be safe and use the default value of 14%. For mounting position, the chosen option will be ‘Roof added / building integrated’. The next part is inputting the degree of slope and degree of azimuth. The slope is referring to its angle towards the sky, while azimuth is referring to the cardinal direction the panel is pointed. The best option on this tool is to check the box that says, “Optimize slope and azimuth.” This gives the best numbers for both. If, for some reason, there is a desire to manually input these values, the best slope is generally around 30 to 45 degrees, and the best azimuth is 0 degrees, pointing straight south toward the equator.

VII Examples of solar carports in the United States

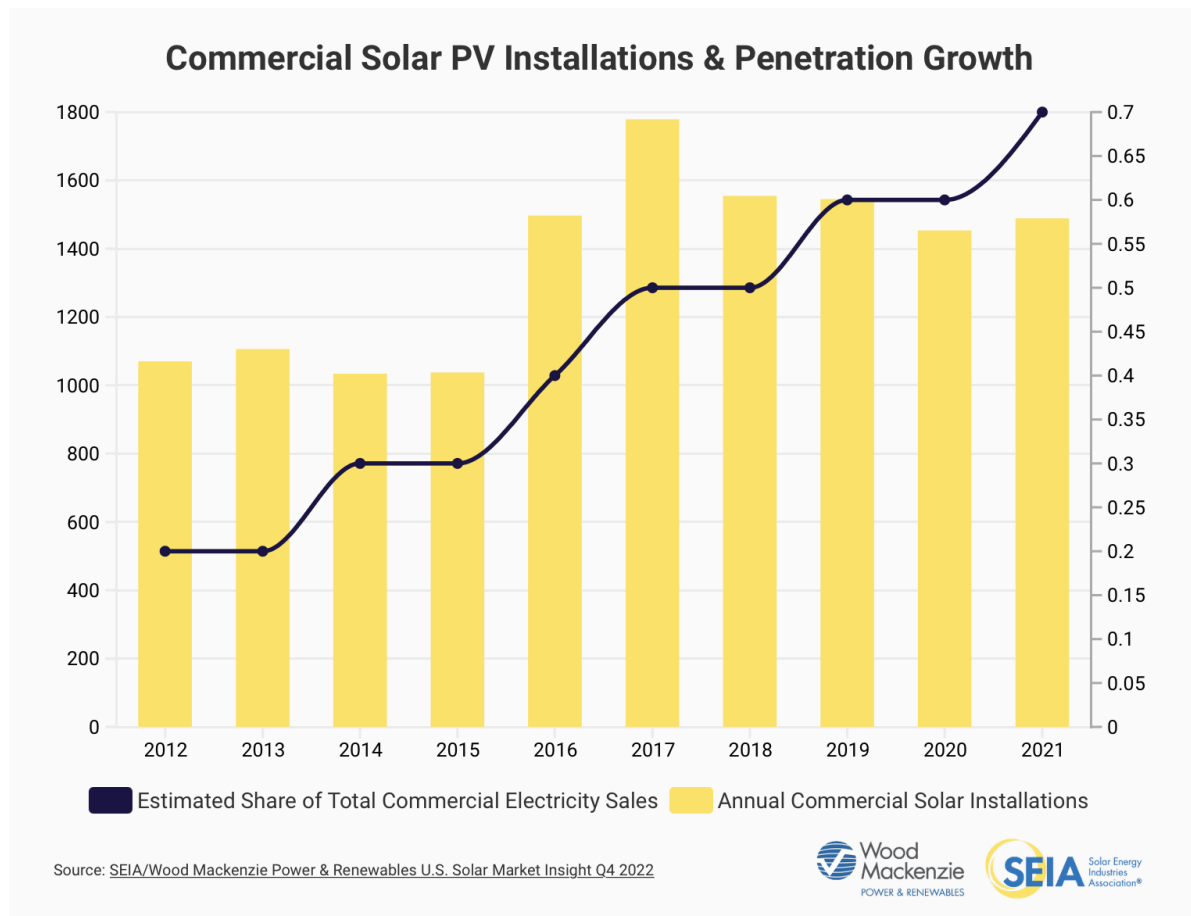
As transportation continues to be one of the main contributors to greenhouse gas emissions in California, the importance of zero-emission vehicles becomes more prominent. Since electric vehicles (EVs) have the potential to significantly reduce emissions, the demand for EVs continues to rise. As a result, companies have searched for new innovative methods to make EVs even more sustainable. One such method is to charge electric vehicles using renewable energy.

Electrify America has installed infrastructure to make all their electric vehicle chargers backed by 100% renewable energy. The company has paved the way for innovative charging methods, “So far, Electrify America has invested \$2 million in 30 solar-powered electric vehicle (EV) charging stations in rural California” (Electrify America). More impressively, these solar-powered EV charging stations do not contribute to the electrical grid. The charging stations instead rely on energy storage, in case the weather is not ideal for the solar panels. These types of solar-powered electric vehicle charging stations are primarily used for Level 2 chargers.

In addition to the 100% solar-powered Level 2 stations, Electrify America has also installed solar canopies that are able to withstand DC fast charging. Unlike the Level 2 solar-powered stations, the solar canopies are connected to the local electrical grid. Regardless, the company has demonstrated the ability to charge electric vehicles by only relying on renewable energy sources. As a result, Electrify America has received several investments and is now valued over \$2.4 billion (Volkswagen Newsroom).

The Ray is another organization of note. It is a non-profit project located in Georgia that is focused on developing and implementing sustainable solutions. The organization became well-known for developing a solar roadway, which is a section of a major highway (I-85) that is capable of generating solar power. The solar roadway is currently just a pilot project, and it aims to test the feasibility and efficiency of generating energy from the road, as well as its potential to power nearby infrastructure. The energy generated by the panels is stored in batteries, which can be used to power nearby buildings, electric vehicles, and other infrastructure. The Ray's solar roadway is still in the testing phase, but it has the potential to be a groundbreaking way of generating renewable energy.

Along the solar roadway, The Ray has installed the state's first ever solar-powered PV4EV charging station. Since these stations have been installed in 2015, they have become a popular addition to the solar roadway. The Ray has partnered with solar and technology companies to further advance the charging stations' power and technology, in 2020. As a result, the newly renovated EV charging stations are now “three times more powerful than the original 50 kW station installed in 2015 with 175 kW Terra HP and modular, high-powered charging system designed to minimize charging time” (The Ray). As more technology continues to develop, The Ray plans to continue renovating the solar roadway and solar-powered EV stations, as well as include “fueling” stations for other alternative-fuel vehicles.



VIII Conclusion

In conclusion, the adoption of solar power in electric vehicle (EV) charging stations and buildings brings numerous benefits that support a sustainable and clean energy future. By leveraging solar energy, we can reduce carbon emissions and improve air quality, thereby addressing climate change and public health concerns. Moreover, solar-powered EV charging stations for buildings as well can increase access to clean energy and promote the widespread adoption of EVs, which are key components of a more sustainable transportation system. Southern California Edison's (SCE) commitment to utilizing renewable energy sources, such as solar power, is an essential step towards a more electric future, and solar power integration in EV stations and buildings will help them reach their goals while alleviating some of the rising electric demands going forward. With the continued adoption of solar energy in the transportation and building sectors, we can build a more sustainable and resilient future for ourselves and future generations.

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